



Dublin and Dún Laoghaire ETB

Programme Module for

Maths for Information Technology

leading to

Level 5 QQI

Maths for Information Technology 5N18396

Introduction

This programme module may be delivered as a standalone module leading to certification in a QQI minor award. It may also be delivered as part of an overall validated programme leading to a Level 5 QQI Certificate.

The purpose of this award is to certify relevant knowledge, skill and competence to apply a broad range of mathematical skills and tools to a wide variety of contexts especially in software development and/or working with computer systems and networks.

The teacher/tutor should familiarise themselves with the information contained in Dublin and Dún Laoghaire ETB's programme descriptor for the relevant validated programme prior to delivering this programme module.

The programme module is structured as follows:

1. Title of Programme Module
2. QQI Component Title and Code
3. Duration in hours
4. Credit Value of QQI Component
5. Status
6. Special Requirements
7. Aim of the Programme Module
8. Objectives of the Programme Module
9. Learning Outcomes
10. Indicative Content
11. Assessment a. Assessment Technique(s) b. Mapping of Learning Outcomes to Assessment Technique(s) c. Guidelines for Assessment Activities
12. Grading
13. Learner Marking Sheet(s), including Assessment Criteria

Integrated Delivery and Assessment

The teacher/tutor is encouraged to integrate the delivery of content where an overlap between content of this programme module and one or more other programme modules is identified. This programme module will facilitate the learner to develop the academic and vocational language, literacy and numeracy skills relevant to the themes and content of the module.

Likewise the teacher/tutor is encouraged to integrate assessment where there is an opportunity to facilitate a learner to produce one piece of assessment evidence which demonstrates the learning outcomes from more than one programme module. The

integration of the delivery and assessment of level 5 Communications and level 5 Mathematics modules with that of other level 5 modules is specifically encouraged, as appropriate.

Indicative Content

The indicative content in Section 10 does not cover all teaching possibilities. The teacher/tutor is encouraged to be creative in devising and implementing other approaches, as appropriate. The use of examples is there to provide suggestions. The teacher/tutor is free to use other examples, as appropriate. The indicative content ensures all learning outcomes are addressed but it may not follow the same sequence as that in which the learning outcomes are listed in Section 9. It is the teacher's/tutor's responsibility to ensure that all learning outcomes are included in the delivery of this programme module.

1. Title of Programme Module Maths for Information Technology
2. Component Name and Code Maths for Information Technology 5N18396
3. Duration in Hours 150 Hours (typical learner effort, to include both directed and self-directed learning).
4. Credit Value 15 Credits.
5. Status This programme module may be compulsory or optional within the context of the validated programme. Please refer to the relevant programme descriptor, Section 9 Programme Structure.
6. Special Requirements None.
7. Aim of the Programme Module The programme module aims to equip the learner with the knowledge, skill and competence to apply a broad range of mathematical skills and tools to a wide variety of contexts, with some theoretical understanding.
8. Objectives of the Programme Module • To broaden and deepen learners' mathematical knowledge and support their use of this knowledge to enrich their lives. • To enable the learner to acquire mathematical skill in problem solving and mathematical investigations in the following areas: basic arithmetic and algebra, set theory & Boolean logic, functions & calculus, geometry & trigonometry, probability & statistics and algorithms & computations. • To assist the learner to develop the academic and vocational language, literacy and numeracy skills related to Maths for Information Technology 5N18396 through the medium of the indicative content. • To enable the learner to take responsibility for his/her own learning. • To facilitate the learner to develop competence in problem solving, mathematical computation, mathematical thinking and conceptual development.

9. Learning Outcomes of Level 5 Maths for Information Technology 5N18396

Learners will be able to:

1. Basic Arithmetic and Algebra

- 1.1. Master the operations of addition, multiplication, subtraction and division in the N, Z, Q, R domains including the rules of indices and logarithms.
- 1.2. Solve practical problems by choosing the correct formulae to calculate the area and perimeter of a square, rectangle, triangle, and circle.
- 1.3. Solve practical problems by choosing the correct formulae, to calculate the volume/capacity and surface area of a cube, cylinder, cone and sphere.
- 1.4. Demonstrate a fundamental understanding of binary numbers. Perform binary arithmetic. Be familiar with hexadecimal notation. Convert from binary and hexadecimal to base 10 and base 10 to binary and hexadecimal.
- 1.5. Distinguish between an expression and an equation.
- 1.6. Evaluate, expand and simplify algebraic expressions.
- 1.7. Transpose formulae and perform arithmetic operations on polynomials and rational algebraic expressions.
- 1.8. Multiply linear expressions to produce quadratics and cubics.
- 1.9. Reduce quadratic expressions to products of linear expressions through the use of inspection to determine the factors. Use this to solve quadratic equations.
- 1.10. Solve quadratic equations with real and complex roots by factorisation or formula.
- 1.11. Solve linear inequalities.
- 1.12. Find a solution, if it exists, for simultaneous linear equations with 2 and 3 unknowns and interpret the results.

2. Set theory and Boolean Logic

- 2.1. Demonstrate a fundamental understanding of the language of set theory, including - universal set, subsets, sets N , Z , Q , R , C and $\emptyset$, finite and infinite sets and cardinal number of a set.
- 2.2. Use the basic operations on sets including union, intersection, complement, symmetric difference, Cartesian product and power set.
- 2.3. Use Venn diagrams of two and three sets to represent relationships between sets.
- 2.4. Define and apply the truth tables for (N) AND, NOT, (N) OR and XOR.
- 2.5. Use truth tables to establish logical equivalences, for example De Morgan's Laws.

3. Functions and Calculus

- 3.1. Recognise that a function assigns a single output to every input, understand the concept of an inverse function and be able to compute it in simple algebraic cases.
- 3.2. Graph linear, quadratic and cubic functions and use these graphs to solve equations $f(x) = 0$, $f(x) = k$ and $f(x) = g(x)$.
- 3.3. Define and graph simple and trigonometric functions.
- 3.4. Complete the square for a quadratic function and hence determine its roots and vertex.
- 3.5. Investigate the concept of the limit of a function and compute the limits of linear, quadratic and quotient functions and understand the idea of a continuous function.
- 3.6. Understand how a derivative arises as a limit from looking for tangent lines or rates of change.
- 3.7. Use the sum, product and quotient formulas for differentiation and the chain rule to differentiate simple functions that are a composition of several functions.

4. **Geometry and Trigonometry**

Co-ordinate geometry

- 4.1. Work with linear equations of the form $ax + by + c = 0$.
- 4.2. Solve problems involving slope of a line to include investigating parallel and perpendicular lines.
- 4.3. Solve problems involving midpoint and length of a line segment.

Trigonometry

- 4.4. Apply the Pythagorean Theorem.
- 4.5. Demonstrate understanding of the concepts of degree and radian measure.
- 4.6. Define $\sin(\theta)$, $\cos(\theta)$ and $\tan(\theta)$, using right angled triangles and using the unit circle having regard to amplitude, phase and period.
- 4.7. Work with trigonometric ratios in root form.
- 4.8. Solve problems involving the area of a triangle using the formula $\text{area} = \frac{1}{2}ab \sin \theta$.
- 4.9. Solve practical problems using trigonometric formulae and terminology, including the sine, cosine and tangent ratios for right angled triangles.
- 4.10. Solve practical problems using the Sine Rule and Cosine Rule.

5. **Probability and Statistics**

Counting

- 5.1. List outcomes of an experiment.
- 5.2. Apply the fundamental principle of counting.

- 5.3. Count the arrangements of n distinct objects ($n!$).
- 5.4. Count the number of ways of arranging r objects from n distinct objects.
- 5.5. Count the number of ways of selecting r objects from n distinct objects.

Probability

- 5.6. Recognise that probability is a measure on a scale of 0-1 of how likely an event is to occur.
- 5.7. Understand the concepts and be able to calculate probabilities by counting equally likely outcomes.
- 5.8. Understand the concepts and be able to calculate compound probabilities of independent events and of mutually exclusive events describing data graphically and numerically.

Statistics

- 5.9. Understand the different types of data: categorical - nominal or ordinal; numerical - discrete or continuous.
- 5.10. Demonstrate an understanding of the relative effectiveness of different displays in representing the findings of a statistical investigation (pie charts, histograms, stem and leaf plots).
- 5.11. Use frequency tables and histograms to display data.
- 5.12. Understand and be able to compute:-
 - mean, median, mode to measure central tendency.
 - range and standard deviation to measure variability.
- 5.13. Discuss the limitations or merits of mean, median and mode for measuring central tendency with symmetric data and with skewed data.

6. Algorithms and Computations

- 6.1. Explain the concept of an algorithm.
- 6.2. Relate the concept of an algorithm to that of a mathematical function acting on a domain of possible inputs.
- 6.3. Manipulate lists and arrays including addition and multiplication.
- 6.4. Use the mathematical notations involved in using lists and arrays including the index, sigma and pi notation.
- 6.5. Apply lists and arrays to simple problems such as shopping lists and prices.
- 6.6. Describe the divide and conquer approach to solving problems.
- 6.7. Use algorithms/functions for carrying out simple operations on one dimensional arrays by iterating over an index.
- 6.8. Use algorithms for simple computational problems using recursion including linear search and binary search; and sort techniques including bubble sort, insertion sort, selection sort and shell sort.

10. Indicative Content

This section provides suggestions for programme content but is not intended to be prescriptive. The programme module can be delivered through classroom based learning activities, group discussions, one-to-one tutorials, field trips, case studies, role play and other suitable activities, as appropriate.

Section 1: Basic Arithmetic and Algebra
1.1. Master the operations of addition, multiplication, subtraction and division in the N, Z, Q, R domains including the rules of indices and logarithms. 1.2. Solve practical problems by choosing the correct formulae to calculate the area and perimeter of a square, rectangle, triangle and circle. 1.3. Solve practical problems by choosing the correct formulae, to calculate the volume/capacity and surface area of a cube, cylinder, cone and sphere. 1.4. Demonstrate a fundamental understanding of binary numbers. Perform binary arithmetic. Be familiar with hexadecimal notation. Convert from binary and hexadecimal to base 10 and base 10 to binary and hexadecimal. 1.5. Distinguish between an expression and an equation. 1.6. Evaluate, expand and simplify algebraic expressions. 1.7. Transpose formulae and perform arithmetic operations on polynomials and rational algebraic expressions. 1.8. Multiply linear expressions to produce quadratics and cubics. 1.9. Reduce quadratic expressions to products of linear expressions through the use of inspection to determine the factors. Use this to solve quadratic equations. 1.10. Solve quadratic equations with real and complex roots by factorisation or formula. 1.11. Solve linear inequalities. 1.12. Find a solution, if it exists, for simultaneous linear equations with 2 and 3 unknowns and interpret the results.
Section 2: Set Theory and Boolean Logic
2.1. Demonstrate a fundamental understanding of the language of set theory including universal set, subsets, sets N, Z, Q, R, C and $\emptyset$, finite and infinite sets, and cardinal number of a set.

- 2.2. Use the basic operations on sets including union, intersection, complement, symmetric difference, Cartesian product, and power set.
- 2.3. Use Venn diagrams of two and three sets to represent relationships between sets.
- 2.4. Define and apply the truth tables for (N) AND, NOT, (N) OR and XOR.
- 2.5. Use truth tables to establish logical equivalences, for example De Morgan's Laws.

Section 3: Functions and Calculus

- 3.1. Recognise that a function assigns a single output to every input, understand the concept of an inverse function and be able to compute it in simple algebraic cases.
- 3.2. Graph linear, quadratic and cubic functions and use these graphs to solve equations $f(x) = 0$, $f(x) = k$, and $f(x) = g(x)$.
- 3.3. Define and graph simple trigonometric functions.
- 3.4. Complete the square for a quadratic function and hence determine its roots and vertex.
- 3.5. Investigate the concept of the limit of a function and compute the limits of linear, quadratic and quotient functions and understand the idea of a continuous function.
- 3.6. Understand how a derivative arises as a limit from looking for tangent lines or rates of change.
- 3.7. Use the sum, product and quotient formulas for differentiation and the chain rule to differentiate simple functions that are a composition of several functions.

Section 4: Geometry and Trigonometry

Co-ordinate geometry

- 4.1. Work with linear equations of the form $ax + by + c = 0$.
- 4.2. Solve problems involving slope of a line to include investigating parallel and perpendicular lines.
- 4.3. Solve problems involving midpoint and length of a line segment.

Trigonometry

- 4.4. Apply the Pythagorean Theorem.
- 4.5. Demonstrate understanding of the concepts of degree and radian measure.

- 4.6. Define $\sin(\theta)$, $\cos(\theta)$ and $\tan(\theta)$, using right angled triangles and using the unit circle having regard to amplitude, phase and period.
- 4.7. Work with trigonometric ratios in root form.
- 4.8. Solve problems involving the area of a triangle using the formula $\text{area} = \frac{1}{2}ab \sin \theta$.
- 4.9. Solve practical problems using trigonometric formulae and terminology, including the sine, cosine and tangent ratios for right angled triangles.
- 4.10. Solve practical problems using the Sine Rule and Cosine Rule.

Section 5: Probability and Statistics

Counting

- 5.1. List outcomes of an experiment.
- 5.2. Apply the fundamental principle of counting.
- 5.3. Count the arrangements of n distinct objects ($n!$).
- 5.4. Count the number of ways of arranging r objects from n distinct objects.
- 5.5. Count the number of ways of selecting r objects from n distinct objects.

Probability

- 5.6. Recognise that probability is a measure on a scale of 0-1 of how likely an event is to occur.
- 5.7. Understand the concepts and be able to calculate probabilities by counting equally likely outcomes.
- 5.8. Understand the concepts and be able to calculate compound probabilities of independent events and of mutually exclusive events describing data graphically and numerically.

Statistics

- 5.9. Understand the different types of data: categorical - nominal or ordinal numerical - discrete or continuous.
- 5.10. Demonstrate an understanding of the relative effectiveness of different displays in representing the findings of a statistical investigation (pie charts, histograms, stem and leaf plots).
- 5.11. Use frequency tables and histograms to display data.
- 5.12. Understand and be able to compute:-

- mean, median and mode to measure central tendency
- range and standard deviation to measure variability

5.13. Discuss the limitations or merits of mean, median and mode for measuring central tendency with symmetric data and with skewed data.

Section 6: Algorithms and Computations

- 6.1. Explain the concept of an algorithm.
- 6.2. Relate the concept of an algorithm to that of a mathematical function acting on a domain of possible inputs.
- 6.3. Manipulate lists and arrays including addition and multiplication.
- 6.4. Use the mathematical notations involved in using lists and arrays including the index, sigma and pi notation.
- 6.5. Apply lists and arrays to simple problems such as shopping lists and prices.
- 6.6. Describe the divide and conquer approach to solving problems.
- 6.7. Use algorithms/functions for carrying out simple operations on one- dimensional arrays by iterating over an index.
- 6.8. Use algorithms for simple computational problems using recursion including linear search and binary search and sort techniques including linear search and binary search; and sort techniques including bubble sort, insertion sort, selection sort and shell sort.

11. Assessment

11a. Assessment Techniques

Assignments (2 X 30%): 60%

Examination Theory: 40%

11b. Mapping of Learning Outcomes to Assessment Techniques

In order to ensure that the learner is facilitated to demonstrate the achievement of all learning outcomes from the component specification, each learning outcome is mapped to an assessment technique(s). This mapping should not restrict an assessor from taking an integrated approach to assessment. All learning outcomes must be assessed in the techniques outlined below.

Learning Outcome		Assessment Technique
Section 1: Basic Arithmetic and Algebra		
1.1	Master the operations of addition, multiplication, subtraction and division in the N, Z, Q, R domains including the rules of indices and logarithms.	Assignment 1 and/or Assignment 2
1.2	Solve practical problems by choosing the correct formulae to calculate the area and perimeter of a square, rectangle, triangle, and circle.	Assignment 1
1.3	Solve practical problems by choosing the correct formulae, to calculate the volume/capacity and surface area of a cube, cylinder, cone, and sphere.	Assignment 1
1.4	Demonstrate a fundamental understanding of binary numbers. Perform binary arithmetic. Be familiar with hexadecimal notation. Convert from binary and hexadecimal to base 10 and base 10 to binary and hexadecimal.	Assignment 1 and/or Assignment 2
1.5	Distinguish between an expression and an equation.	Assignment 1 and/or Assignment 2
1.6	Evaluate, expand and simplify algebraic expressions.	Assignment 1 and/or Assignment 2
1.7	Transpose formulae and perform arithmetic operations on polynomials and rational algebraic expressions.	Assignment 1 and/or Assignment 2
1.8	Multiply linear expressions to produce quadratics and cubics.	Assignment 1 and/or Assignment 2
1.9	Reduce quadratic expressions to products of linear	Assignment 1 and/or

	expressions through the use of inspection to determine the factors. Use this to solve quadratic equations.	Assignment 2
1.10	Solve quadratic equations with real and complex roots by factorisation or formula.	Assignment 1 and/or Assignment 2
1.11	Solve linear inequalities.	Assignment 1 and/or Assignment 2
1.12	Find a solution, if it exists, for simultaneous linear equations with 2 and 3 unknowns and interpret the results.	Assignment 1 and/or Assignment 2
Section 2: Set Theory and Boolean Logic		
2.1	Demonstrate a fundamental understanding of the language of set theory including universal set, subsets, sets N , Z , Q , R , C and $\emptyset$, finite and infinite sets, and cardinal number of a set.	Assignment 1 and/or Assignment 2 and Exam
2.2	Use the basic operations on sets including union, intersection, complement, symmetric difference, Cartesian product and power set.	Assignment 1 and/or Assignment 2 and Exam
2.3	Use Venn diagrams of two and three sets to represent relationships between sets.	Assignment 1 and/or Assignment 2 and Exam
2.4	Define and apply the truth tables for (N) AND, NOT, (N) OR and XOR.	Assignment 1 and/or Assignment 2
2.5	Use truth tables to establish logical equivalences, for example De Morgan's Laws.	Assignment 1 and/or Assignment 2
Section 3: Functions and Calculus		
3.1	Recognise that a function assigns a single output to every input, understand the concept of an inverse function and be able to compute it in simple algebraic cases.	Exam and/or Assignment 2
3.2	Graph linear, quadratic and cubic functions and use these graphs to solve equations $f(x) = 0$, $f(x) = k$, and $f(x) = g(x)$.	Exam
3.3	Define and graph simple trigonometric functions.	Exam
3.4	Complete the square for a quadratic function and hence determine its roots and vertex.	Exam
3.5	Investigate the concept of the limit of a function and compute the limits of linear, quadratic and quotient functions and understand the idea of a continuous function.	Exam

3.6	Understand how a derivative arises as a limit from looking for tangent lines or rates of change.	Exam
3.7.	Use the sum, product and quotient formulas for differentiation and the chain rule to differentiate simple functions that are a composition of several functions.	Exam
Section 4: Geometry and Trigonometry		
Co-ordinate Geometry		
4.1	Work with linear equations of the form $ax + by + c = 0$	Exam
4.2	Solve problems involving slope of a line to include investigating parallel and perpendicular lines	Exam
4.3	Solve problems involving midpoint and length of a line segment	Exam
Trigonometry		
4.4	Apply the Pythagorean theorem.	Exam
4.5	Demonstrate understanding of the concepts of degree and radian measure.	Exam
4.6	Define $\sin(\theta)$, $\cos(\theta)$ and $\tan(\theta)$, using right angled triangles and using the unit circle having regard to amplitude, phase and period.	Exam
4.7	Work with trigonometric ratios in root form.	Exam
4.8	Solve problems involving the area of a triangle using the formula $\text{area} = \frac{1}{2}ab \sin \theta$.	Exam
4.9	Solve practical problems using trigonometric formulae and terminology, including the sine, cosine and tangent ratios for right angled triangles.	Exam
4.10	Solve practical problems using the Sine Rule and Cosine Rule.	Exam
Section 5: Probability and Statistics		
Counting		
5.1	List outcomes of an experiment.	Assignment 1 and/or Assignment 2 and Exam
5.2	Apply the fundamental principle of counting.	Assignment 1 and Exam

5.3	Count the arrangements of n distinct objects ($n!$).	Assignment 1 and Exam
5.4	Count the number of ways of arranging r objects from n distinct objects.	Assignment 1 and Exam
5.5	Count the number of ways of selecting r objects from n distinct objects.	Assignment 1 and Exam
Probability		
5.6	Recognise that probability is a measure on a scale of 0-1 of how likely an event is to occur.	Assignment 1 and Exam
5.7	Understand the concepts and be able to calculate probabilities by counting equally likely outcomes.	Assignment 1 and Exam
5.8	Understand the concepts and be able to calculate compound probabilities of independent events and of mutually exclusive events describing data graphically and numerically.	Assignment 1 and Exam
Statistics		
5.9	Understand the different types of data: categorical - nominal or ordinal numerical - discrete or continuous.	Assignment 1 and Exam
5.10	Demonstrate an understanding of the relative effectiveness of different displays in representing the findings of a statistical investigation (pie charts, histograms, stem and leaf plots).	Assignment 1 and/or Assignment 2 and Exam
5.11	Use frequency tables and histograms to display data.	Assignment 1 and/or Assignment 2 and Exam
5.12	Understand and be able to compute:- • mean, median and mode to measure central tendency • range and standard deviation to measure variability.	Assignment 1 and Exam
5.13	Discuss the limitations or merits of mean, median and mode for measuring central tendency with symmetric data and with skewed data.	Assignment 1 and Exam
Section 6: Algorithms and Computations		
6.1	Explain the concept of an algorithm.	Assignment 2
6.2	Relate the concept of an algorithm to that of a mathematical function acting on a domain of possible inputs.	Assignment 2

6.3.	Manipulate lists and arrays including addition and multiplication.	Assignment 2
6.4	Use the mathematical notations involved in using lists and arrays including the index, sigma and pi notation.	Assignment 2
6.5	Apply lists and arrays to simple problems such as shopping lists and prices.	Assignment 2
6.6	Describe the divide and conquer approach to solving problems.	Assignment 2
6.7	Use algorithms/functions for carrying out simple operations on one-dimensional arrays by iterating over an index.	Assignment 2
6.8	Use algorithms for simple computational problems using recursion including linear search and binary search and sort techniques including bubble sort, insertion sort, selection sort and shell sort.	Assignment 2

11c. Guidelines for Assessment Activities

The assessor is required to devise assessment briefs/examination paper and marking schemes/outline solutions for the assignments and examination. In devising the assessment briefs/examination paper, care should be taken to ensure that the learner is given the opportunity to **show evidence of achievement of ALL the learning outcomes**. Assessment briefs may be designed to allow the learner to make use of a wide range of media in presenting assessment evidence, as appropriate. Quality assured procedures must be in place to ensure the reliability of learner evidence.

Assessment technique(s) including weighting(s)

In order to demonstrate that they have reached the standards of knowledge, skill and competence identified in all the learning outcomes, learners are required to complete the assessment(s) below.

Assignments (2)	60% (2 x 30%)
Assignment 1 - 30%	
Four weeks preparation period where the course is taught over an academic year or appropriate pro rata time period if the course is taught over a different duration. The requirements of this assignment can be met by either a project or a proctored assessment approach.	
The brief for the first assignment will cover learning outcomes from Basic Arithmetic & Algebra, Set Theory & Boolean Logic and Probability & Statistics.	
Assignment 2 - 30%	
Four weeks preparation period where the course is taught over an academic year or appropriate pro rata time period if the course is taught over a different duration	
The assessor will devise a brief and marking criteria to cover the learning outcomes from Section 1, 2, 3, 5 & 6. This section involves the application of the relevant mathematical principles to a technological vocational area, e.g. queuing theory in networking, programming in software development or system design in electronics. The assignment brief should examine the student's knowledge of the application of Algorithms and Computations methods appropriate to the program being followed.	

Examination	40%
2 Hours	
The examination (120 Marks - 40%) will be based on a range of learning outcomes from sections 2, 3, 4 and 5 of 2 hours duration. The format of the exam will be as follows: Section A – 40 Marks 12 short answer questions covering all sections (4 Marks each). Candidates are required to answer any ten questions in this section. Section B – 80 Marks 6 structured questions (20 Marks each). Candidates are required to answer four questions from this section.	

12. Grading

Distinction: 80% - 100%

Merit: 65% - 79%

Pass: 50% - 64%

Unsuccessful: 0% - 49%

At levels 4, 5 and 6 major and minor awards will be graded. The grade achieved for the major award will be determined by the grades achieved in the minor awards.

Maths for Information Technology 5N18396	Learner Marking Sheet Assignments 60% (2 X 30%)
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Learner's Name: _____

Assessment Criteria	Maximum Mark	Learner Mark
Assignment 1 The following assessment criteria will be reflected in the individual marking scheme. • Effective application of mathematics to real life situations. • Use of appropriate mathematics to verify and interpret results. • Accurate calculations and correct use of formulae. • Coherent format with appropriate use of mathematical symbols, letters and terminology. • Logical progression of thought.	12 12 12 12 12	
Subtotal	60	
Subtotal No.1 ÷ 2	30%	
Assignment 2 The following assessment criteria will be reflected in the individual marking scheme. • Outline of the solution to the problem specified is appropriate • Logical progression of thought • Effective application of algorithms and computational methods • Coherent format with appropriate use of mathematical symbols, letters and terminology • The solution is tested and verified	12 12 12 12 12	
Subtotal	60	
Subtotal No 2 ÷ 2	30%	
Total Marks = Subtotal 1 (%) + Subtotal 2 (%)	60%	

Assessor's Signature: _____

Date: _____

External Authenticator's Signature: _____

Date: _____

Maths for Information Technology 5N18396	Learner Marking Sheet Examination (40%)
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Learner's Name: _____

Assessment Criteria	Maximum Mark	Learner Mark
Section A: Twelve Short Answer Questions Answer any ten questions. Question No.: _____ _____ _____ _____ _____ _____ _____ _____ _____ _____	 4 4 4 4 4 4 4 4 4 4	
Subtotal 1:	40	
Section B: Six Structured Questions Answer any four questions. Question No.: _____ Question No.: _____ Question No.: _____ Question No.: _____	 20 20 20 20	
Subtotal 2:	80	
Total Marks = (Subtotal 1 + Subtotal 2) ÷ 3	40%	

Assessor's Signature: _____ Date: _____

External Authenticator's Signature: _____ Date: _____